

BINARY ARITHMETIC

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BINARY ARITHMETIC

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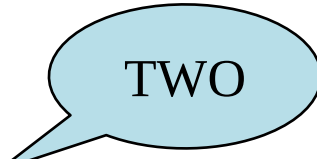
BINARY ARITHMETIC

BINARY ADDITION:

The rule for addition of binary numbers are

A	B	SUM	CARRY
Augends	Addend	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

A	B	A + B
0	0	0
0	1	1
1	0	1
1	1	10



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EXAMPLE:

	MSB		LSB		Decimal
<i>Carry</i> →	1	1	1	0	
	1	1	1	1	15
+	1	0	1	0	10
	1	1	0	0	25

PROCEDURE: Step 1: The Least Significant Bits are added. $0 + 1 = 1$ with carry 0.

Step 2: The carry in the previous step is added to the next higher significant bits, i.e.

$0 + 1 + 1 = 0$ with carry 1.

Step 3: The carry in the above step is added to the next higher significant bits, i.e.

$1 + 1 + 0 = 0$ with carry 1.

Step 4: The preceding carry is added to the next higher significant bits, i.e.

$1 + 1 + 1 = 0$ with carry 1.

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EXAMPLE:

Add 1011.011 and 110.1

1 0 1 1 . 0 1 1	=	11.375
0 1 1 0 . 1 0 0	=	6.5
1 0 0 0 1 . 1 1 1	=	17.875

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BINARY SUBTRACTION:

A	B	DIFFERENCE	BORANE
Minuend	Subtrahend	D	B
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

NOTE: When the digit 1 is subtracted from zero, a borrow of 1 is taken from left most digits which forms 10 in binary (2 in decimal number), so if 1 is subtracted from 10 answer is 1 (i.e. $2-1 = 1$).

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EXAMPLE: Subtract 100_2 from 1001_2

$$\begin{array}{r} 0 \quad 10 \quad \leftarrow \text{Borrow} \\ \cancel{1} \quad \cancel{0} \quad 0 \quad 1 \\ \quad 1 \quad 0 \quad 0 \\ \hline \quad 1 \quad 0 \quad 1 \\ \hline \end{array}$$

Answer: $100_2 - 1001_2 = 101_2$

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EXAMPLE: Subtract by using binary arithmetic 7.25 from 9.50

$$\begin{array}{r} \text{1} \\ \text{10 10} \quad \text{0 10} \\ \text{1 0 0 1 . 1 0} \\ - \text{0 1 1 1 . 0 1} \\ \hline \text{0 0 1 0 . 0 1} \end{array} \qquad \begin{array}{r} 9.50 \\ -7.25 \\ \hline 2.25 \end{array}$$

Exercise: Subtract by using binary arithmetic 6.25 from 10.75

Answer: (100.10)₂

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BINARY MULTIPLICATION:

A	B	$A \times B$
0	0	0
0	1	0
1	0	0
1	1	1

PROCEDURE: **Step 1:** The LSB of multiplier is taken. If the multiplier bit is 1, the multiplicand is copied as such and, if the multiplier bit is 0, a 0 is placed in all the bit position.

Step 2: The next higher significant bit of the multiplier is taken and the partial product is written with a shift to the left, as in step 1.

Step 3: Step 2 is repeated for all other higher significant bits and each time a left shift is given.

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step 4: When all the bits in the multiplier have been taken into account, the partial product terms are added, which give the actual product of the multiplier and the multiplicand.

EXAMPLE: Multiply 1011 by 1101

$$\begin{array}{r} 1011 \\ 1101 \\ \hline 1011 \\ 0000 \\ 1011 \\ 1011 \\ \hline 10001111 \\ \hline \end{array}$$

BINARY ARITHMETIC

BINARY DIVISION: Division in binary follows the same procedure as division in decimal. Division by 0 is meaningless. Example

$$11001 \div 101$$

$$\begin{array}{r} 101 \overline{) 11001} (101 \\ \underline{101} \\ 0101 \\ \underline{101} \\ 0000 \end{array}$$

THANK YOU

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